Session 7 Homework — Scoreboard & Strategy 3

Kacper Dudzik · MILK10k dermoscopic classification (macro-F1)

https://github.com/KacperDudzik/CV_session
Code: session7/homework.py · session7/session7.py · homework_results.json

1. Scoreboard (Sessions 5–7)

Metric: lesion-level **macro-F1**. Session 7 rows were trained in this homework run; test is scored once after the best validation epoch.

Model	Val macro-F1	Test macro-F1
Session 5 - random forest (33 features, course)	0.534	0.495
Session 5+ midterm SVM (all_features, oversample)	0.567	0.597
Session 6 - MLP 33-64-3 (33 features)	0.493	0.472
Session 7 - SimpleCNN (scratch)	0.522	0.510
Session 7 - ResNet-18 frozen + linear head (strategy 1)	0.629	0.560
Session 7 - ResNet-18 fine-tune layer4+fc (strategy 2)	0.645	0.613
Session 7 - ResNet-18 fine-tune layer3+layer4+fc	0.660	0.581
Session 7 - ResNet-18 full fine-tune (strategy 3)	0.644	0.564

Best model: Session 7 - ResNet-18 fine-tune layer4+fc (strategy 2) — val 0.645, **test macro-F1 = 0.613**.

2. DenseNet vs ResNet (one sentence)

DenseNet differs from ResNet by concatenating each layer's feature maps to all later layers (dense connectivity) instead of adding residual skip connections, which encourages feature reuse and typically needs fewer parameters.

3. Optional: full fine-tuning (strategy 3)

Strategy 3 unfreezes every ResNet-18 parameter (≈11.2M trainable) and trains at lr=1e-4 with class-weighted CE and flips, same protocol as partial fine-tuning.

Setting	Trainable params	s/epoch	Val F1	Test F1
Strategy 2 (layer4+fc)	8,395,267	94	0.645	0.613
Strategy 3 (full network)	11,178,051	161	0.644	0.564

Finding: Full fine-tuning did not beat partial layer4 fine-tuning on test; keep strategy 2 as default.

Notes

- Session 7 models use lesion-grouped dermoscopic StratifiedGroupKFold (seed=0).
- Midterm SVM uses its own seed=42 lesion split; reported for completeness as best classical.
- Test macro-F1 is computed once per model after selecting the best validation checkpoint.

Device: cpu · seed 0 · splits {'train': 3668, 'val': 1048, 'test': 524}